

The Silent Tax: Typo-Robustness Analysis (GSM8K_FIX-WARN)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. `n` = questions; `n_answered` = questions that produced a final answer (not truncated at the token cap). `acc_answered` = correct / `n_answered`; `acc_strict` = correct / `n` (truncated counted as wrong); `ci_lo/ci_hi` = 95% bootstrap CI on `acc_strict`.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	499	0.9238	0.922	0.898	0.944
typo25_real10	500	498	0.8574	0.854	0.822	0.884
typo25_real40	500	499	0.8717	0.87	0.84	0.898
typo25_real70	500	496	0.8548	0.848	0.816	0.878
typo50_real10	500	491	0.7943	0.78	0.746	0.814
typo50_real40	500	487	0.7474	0.728	0.692	0.766
typo50_real70	500	497	0.7404	0.736	0.696	0.774
typo75_real10	500	496	0.7056	0.7	0.66	0.742
typo75_real40	500	481	0.6632	0.638	0.596	0.68
typo75_real70	500	477	0.6394	0.61	0.566	0.654

Table 2: Answered-only flips vs clean, paired by question over `n_both` (answered in both conditions). `r2w` / `w2r` = questions correct on clean but wrong on this config / wrong on clean but correct on this config; `r2w_ratio` = `r2w` / `n_both`, `w2r_ratio` = `w2r` / `n_both`; `net_flip_ratio` = `r2w_ratio` - `w2r_ratio` = (`r2w` - `w2r`) / `n_both`, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	497	47	14	0.0946	0.0282	0.0664	47	14	16.787	4.182e-05
typo25_real40	498	40	14	0.0803	0.0281	0.0522	40	14	11.574	0.00066878
typo25_real70	495	49	15	0.099	0.0303	0.0687	49	15	17.016	3.707e-05
typo50_real10	490	78	14	0.1592	0.0286	0.1306	78	14	43.141	0.0
typo50_real40	486	101	15	0.2078	0.0309	0.177	101	15	62.284	0.0
typo50_real70	496	101	10	0.2036	0.0202	0.1835	101	10	72.973	0.0
typo75_real10	495	124	17	0.2505	0.0343	0.2162	124	17	79.688	0.0
typo75_real40	480	137	13	0.2854	0.0271	0.2583	137	13	100.86	0.0
typo75_real70	476	146	13	0.3067	0.0273	0.2794	146	13	109.585	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	499	1475.9	1178	2429	14374
typo25_real10	498	1747.6	1189.0	3521	15526
typo25_real40	499	1622.3	1179	2970	12683
typo25_real70	496	1693.5	1190.0	3548	12899
typo50_real10	491	1813.7	1211	3824	19443
typo50_real40	487	1791.4	1170	3887	20000
typo50_real70	497	2011.8	1227	4958	17142
typo75_real10	496	2328.2	1429.0	5507	12967
typo75_real40	481	2172.5	1342	5114	14785
typo75_real70	477	2371.8	1412	5688	12236

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1338.0	3148.5	2.3532
typo25_real10	1425.5	3684.4	2.5845
typo25_real40	1411.3	3056.1	2.1654
typo25_real70	1402.4	3407.2	2.4294
typo50_real10	1417.8	3342.8	2.3578
typo50_real40	1413.4	2910.1	2.0589
typo50_real70	1474.5	3544.6	2.4039
typo75_real10	1765.3	3677.7	2.0834
typo75_real40	1595.2	3309.3	2.0745
typo75_real70	1733.1	3504.4	2.0221

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	499	13.507	10.956	7.674	6.224	21.181	17.18
typo25_real10	498	14.388	14.106	13.3	13.04	27.688	27.147
typo25_real40	499	14.023	12.673	12.36	11.17	26.384	23.844
typo25_real70	496	14.386	13.827	13.172	12.659	27.558	26.486
typo50_real10	491	14.209	14.365	15.148	15.314	29.358	29.678
typo50_real40	487	14.039	13.793	15.671	15.396	29.71	29.189
typo50_real70	497	14.454	16.569	17.617	20.195	32.072	36.765
typo75_real10	496	14.431	19.248	20.78	27.716	35.211	46.964
typo75_real40	481	14.282	17.561	21.227	26.102	35.509	43.663
typo75_real70	477	14.072	19.421	23.479	32.405	37.551	51.826

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	3.555	9.349	5.793
wait	second_guess	6.199	8.477	2.278
or maybe/alt	uncertainty	2.127	3.894	1.767
perhaps	uncertainty	0.877	1.934	1.057
unclear/confused	uncertainty	0.546	1.233	0.687
might be	uncertainty	0.104	0.398	0.295
not sure	uncertainty	0.037	0.266	0.229
could be	uncertainty	0.138	0.192	0.053
I assume	uncertainty	0.022	0.06	0.038
not entirely	uncertainty	0.022	0.054	0.032
possibly	uncertainty	0.002	0.008	0.006
on second thought	second_guess	0.0	0.0	0.0
presumably	uncertainty	0.01	0.01	-0.0
reconsider	second_guess	0.002	0.001	-0.002
wait/actually no	second_guess	0.393	0.378	-0.015
but wait	second_guess	0.704	0.678	-0.027
actually	second_guess	0.388	0.31	-0.078
I guess	uncertainty	0.232	0.083	-0.149
hold on	second_guess	0.796	0.599	-0.196
let me recheck	second_guess	0.929	0.539	-0.39
hmm	second_guess	3.111	2.71	-0.4
double-check	second_guess	0.986	0.567	-0.418

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	13.026	15.8	1.213	6.163	14.873	2.413	19.189	30.673	1.598
typo25_real10	14.029	15.163	1.081	10.328	19.719	1.909	24.357	34.882	1.432
typo25_real40	13.599	15.264	1.122	10.16	18.784	1.849	23.759	34.048	1.433
typo25_real70	13.629	16.05	1.178	9.762	20.664	2.117	23.391	36.714	1.57
typo50_real10	13.833	14.791	1.069	11.53	20.738	1.799	25.363	35.529	1.401
typo50_real40	13.835	14.326	1.035	12.058	20.754	1.721	25.893	35.08	1.355
typo50_real70	13.723	15.259	1.112	13.107	22.581	1.723	26.83	37.841	1.41
typo75_real10	14.055	14.836	1.055	16.855	25.002	1.483	30.911	39.837	1.289
typo75_real40	14.171	14.381	1.015	17.433	24.642	1.413	31.604	39.023	1.235
typo75_real70	14.065	14.077	1.001	19.126	27.078	1.416	33.192	41.155	1.24

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3664	2554	555	133	422	0.6971	0.1515	0.0363	0.1152
typo25_real40	3668	2432	632	163	441	0.663	0.1723	0.0444	0.1202
typo25_real70	3639	2265	776	182	416	0.6224	0.2132	0.05	0.1143
typo50_real10	7079	4299	1584	343	853	0.6073	0.2238	0.0485	0.1205
typo50_real40	6995	4011	1675	401	908	0.5734	0.2395	0.0573	0.1298
typo50_real70	7136	3723	1999	519	895	0.5217	0.2801	0.0727	0.1254
typo75_real10	9967	4616	3493	784	1074	0.4631	0.3505	0.0787	0.1078
typo75_real40	9634	4438	3194	840	1162	0.4607	0.3315	0.0872	0.1206
typo75_real70	9530	3828	3538	1013	1151	0.4017	0.3712	0.1063	0.1208

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* $\rightarrow$ *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns *_n are counts; *_frac are the share of n_corrupt_words.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	20710	0.5538	0.2719	0.0608	0.1134
40	20297	0.5361	0.271	0.0692	0.1237
70	20305	0.4834	0.3109	0.0844	0.1213

Table 10: Misread rate split by final correctness. A typo'd word is 'misread' when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo'd words that were misread) / (all typo'd words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0342	0.0476	1.391
typo25_real40	0.0396	0.0724	1.826
typo25_real70	0.0457	0.0736	1.609
typo50_real10	0.0447	0.0615	1.377
typo50_real40	0.0553	0.0626	1.131
typo50_real70	0.0642	0.0942	1.468
typo75_real10	0.0675	0.103	1.526
typo75_real40	0.0792	0.1021	1.29
typo75_real70	0.0898	0.131	1.458

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset's result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. $\text{acc_low_real} / \text{acc_high_real}$ = accuracy on the lower / higher real-word variant; $\text{delta_acc} = \text{acc_high_real} - \text{acc_low_real}$; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with $\text{r2w} \neq \text{w2r}$ means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	494	0.8563	0.8583	0.002	33	34	1.0
25	real10_to_real40	497	0.8571	0.8732	0.0161	30	38	0.39595103
25	real40_to_real70	495	0.8727	0.8545	-0.0182	34	25	0.29763832
50	real10_to_real70	489	0.7935	0.7505	-0.0429	71	50	0.06903635
50	real10_to_real40	481	0.7963	0.7505	-0.0457	69	47	0.05119984
50	real40_to_real70	484	0.7459	0.7479	0.0021	48	49	1.0
75	real10_to_real70	474	0.7068	0.6435	-0.0633	82	52	0.01223758
75	real10_to_real40	477	0.7149	0.6646	-0.0503	81	57	0.05024292
75	real40_to_real70	461	0.6681	0.6464	-0.0217	63	53	0.40336364

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_^* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_}^* = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4422	-0.06385	-0.0405	0.9381	0.9603	1.98872

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	318	0.7233	0.2767
40	349	0.6934	0.3066
70	373	0.756	0.244

Lexical Grid (overview)

Table 14: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	13.507	7.674	1.9	1.3
typo25_real10	14.388	13.3	4.008	2.495
typo25_real40	14.023	12.36	4.26	2.772
typo25_real70	14.386	13.172	4.296	2.872
typo50_real10	14.209	15.148	4.757	3.189
typo50_real40	14.039	15.671	4.839	3.229
typo50_real70	14.454	17.617	5.143	3.704
typo75_real10	14.431	20.78	5.201	3.484
typo75_real40	14.282	21.227	5.243	3.818
typo75_real70	14.072	23.479	5.585	3.972

Spellcheck Recovery